

(Implementation of SLL)

```
#include <stdio.h>
#include <conio.h>
#include <stdlib.h>
```

```
void main()
```

```
{
```

```
    struct node
```

```
    {
```

```
        int n;
```

```
        struct node *ptr;
```

```
    };
```

```
    typedef struct node NODE;
```

```
    NODE *first, *temp, *head;
```

```
    int count = 0;
```

```
    char choice;
```

```
    first = NULL;
```

```
do
```

```
{
```

```
    head = (NODE *) malloc(sizeof(NODE));
```

```
    printf("Enter the data item\n");
```

```
    scanf("%d", &head->n);
```

```
    if (first != NULL)
```

```
    {
```

```
        temp->ptr = head;
```

```
        temp = head
```

```
    }
```

```
    else
```

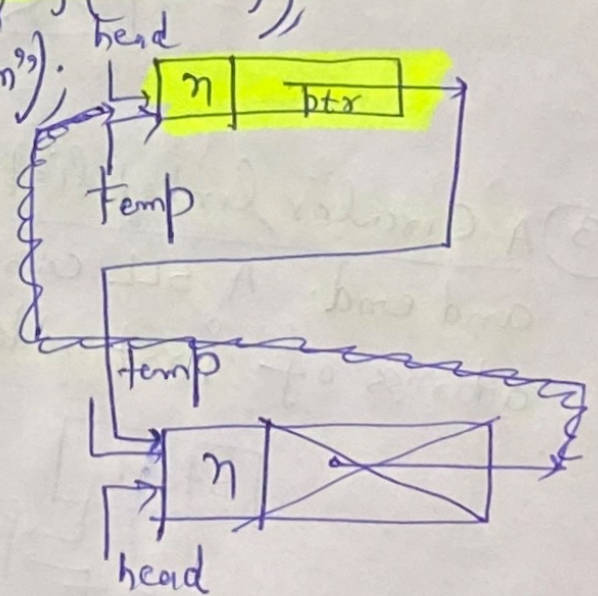
```
    {
```

```
        first = temp = head;
```

```
    }
```

```
    while ((choice == 'Y') || (choice == 'y'));
```

```
    temp->ptr = NULL;
```



```

temp = first;
printf("The linked list is: \n");
while (temp != NULL)
{
    printf("%d \n", temp->num);
    count++;
    temp = temp->ptr;
}
}

```

Insertion in SLL

Insertion a new node into SLL, has following cases:

- Insertion at beginning of SLL.
- Insertion at end of SLL.
- Insertion at specified position with SLL.

In order to insert a new node at one of the above three situations, we need to find position of insertion:

- IF SLL is empty or node to be inserted before the starting node, then insert node at the beginning of SLL as starting node.
- if node to be inserted after the last node in SLL then insert at the end of SLL.
- if above two conditions don't hold true then insert the new node at the specified position in SLL.

Let us see pseudo code of each case:

```
void insert-begin(int item)
```

```
{
```

```
    NODE *p;
```

```
    p = (NODE *) malloc(sizeof(NODE));
```

```
    p->info = item;
```

```
    if(start == NULL)
```

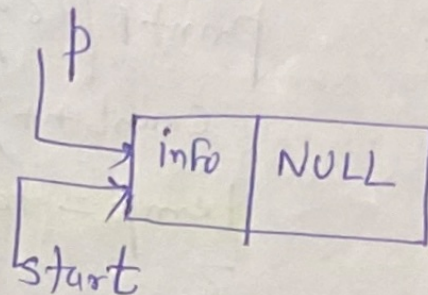
```
        p->next = NULL
```

```
    else
```

```
        p->next = start;
```

```
        start = p;
```

```
}
```



```
void insert-end(int item)
```

```
{
```

```
    NODE *p, *loc;
```

```
    p = (NODE *) malloc(sizeof(NODE));
```

```
    p->info = item;
```

```
    p->next = NULL;
```

```
    if(start == NULL)
```

```
        start = p;
```

```
    else
```

```
        loc = start;
```

```
        while(loc->next != NULL)
```

```
            loc = loc->next;
```

```
        loc->next = p;
```

```
}
```

```
}
```

(4)

```
void insert-spec-loc (int item, int loc)
```

```
{
    NODE *p, *temp;
    int k;
    for (k=0; temp=start; k<loc; k++)
    {
        temp = temp->next;
    }
    if (temp == NULL)
    {
        // k is less than
        printf("NODE at first location");
        return;
    }
}
```

```
{
    p = (NODE*) malloc (sizeof(NODE));
    p->info = item;
    p->next = loc->next;
    loc->next = p;
}
```

